

AHMAD FIRAS AHMAD FER ROUSE

419-770-5814 | firas.azfar@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#) | Toledo, OH

EDUCATION

The University of Toledo

Toledo, OH

Bachelor of Science in Computer Science & Engineering

- GPA: 3.23 | Graduated: May 2026
- Relevant Coursework: Machine Learning, Artificial Neural Networks, Database Management Systems, Software Engineering, Data Structures & Algorithms, Engineering Statistics, Embedded Systems, Computer Architecture, Theory of Computation, Operating Systems & Systems Programming

PROFESSIONAL EXPERIENCE

Microsoft Solution Developer (Internship)

Jan 2026 – May 2026

First Solar — Perrysburg, OH

- Design and build agentic AI solutions in Microsoft Copilot Studio, translating business requirements into autonomous multi-step workflows using custom connectors, Power Automate flows, and Azure AI services to enable intelligent automation across Microsoft 365 and enterprise systems
- Developed, tested, and debugged automated business process workflows end-to-end — conducting unit and system-level testing to ensure robust, scalable, and compliant solutions meeting stakeholder requirements and governance standards
- Led a team of two interns on agentic AI projects using Microsoft Copilot Studio, coordinating task delegation, code reviews, and milestone delivery while maintaining enterprise compliance policies
- Authored comprehensive technical documentation and governance artifacts for all deployed solutions, ensuring maintainability, auditability, and smooth handoff to stakeholders

Sports Analytics & Data Science Intern

Oct 2025 – Present

The University of Toledo Athletic Department — Toledo, OH

- Develop ML-powered statistical models in Python and scikit-learn to evaluate roster efficiency and optimize student-athlete benefit allocation; build interactive DOMO dashboards translating complex performance data into actionable coaching insights
- Collaborate with Athletics Directors and Coaches to implement data-driven decision frameworks; collect, clean, and analyze large game-statistics datasets to surface recommendations supporting coaching and game planning
- Facilitate onboarding and technical training for incoming interns, maintaining departmental performance standards and knowledge continuity

Undergraduate Researcher (Research Co-Op, USRCAP Fellow — \$3,000)

May 2025 – Dec 2025

Learning & Optimization in Networks & Graphs Lab | Cyber-Physical-Human Systems Lab — The University of Toledo

Interactive Graph-Based Reinforcement Learning for UAV Autonomy

- Achieved 72.8% zero-shot UAV deployment success (vs. 43.2% ORB-SLAM3 and 51.7% FlightGoggles) using MAML across 1,000+ simulated environments
- Engineered Human-in-the-Loop control system achieving 97.3% HITL intervention success with sub-100ms switching latency and 84% human trust calibration across three interaction modes
- Built multi-model AI architecture (LLM, VLM, SAM, LAM, LCM) with edge-first intelligence: 95% local processing and sub-100ms inference, eliminating cloud dependencies for GPS-denied environments
- Developed Deep Flyer — an AWS DeepRacer-inspired UAV training platform using P3O reinforcement learning and YOLO11 stereo vision, enabling student-accessible UAV AI research

Senior Design Project — Team Leader

Aug 2024 – May 2025

Homeowner Loss History Prediction | Grange Insurance Partnership — The University of Toledo

- Led interdisciplinary team across EECS, Mathematics & Statistics, and Actuarial Sciences in industry partnership with Grange Insurance, delivering an MLOps-automated insurance risk model achieving $R^2 = 0.976$ and 6% improvement over raw-attribute baseline (~\$450 reduction in pricing error)
- Architected Airflow + MLflow pipeline with XGBoost, AI drift-monitoring agents, and self-healing HITL validation, reducing manual intervention by 75% through comprehensive automation workflows
- Designed real-time HITL oversight dashboard with automated model retraining triggers, hyperparameter optimization, and AWS S3 data ingestion; comprehensive evaluation using RMSE/MAE/MSE metrics
- Delivered production-ready system with stakeholder-friendly visualization dashboards enabling actionable insurance risk assessment; presented findings to Grange Insurance partners and faculty advisors

Database Engineer Co-Op

May 2023 – Aug 2023

Park Place Technologies — Toledo, OH

- Architected centralized PostgreSQL security database consolidating data from Active Directory, Cisco AMP, and Microsoft Defender via a data governance framework — accelerating incident response by 15% and improving data integrity by 95% across all security reporting workflows
- Designed and implemented comprehensive validation rules ensuring strict regulatory adherence; revamped disaster recovery protocols and executed quarterly simulations achieving zero data loss
- Authored detailed technical documentation for the Security Discovery Tool, supporting maintainability and compliance; collaborated with senior engineers to align implementation with organizational security posture requirements

IT Student Assistant

Oct 2022 – Mar 2023

The University of Toledo — Toledo, OH

- Streamlined account troubleshooting processes, resolving 60% of recurring IT issues for students and medical staff; authored detailed technical guides reducing repeat requests by 40%
- Ensured 100% compliance with FERPA and HIPAA guidelines while handling sensitive student and patient data; provided cross-platform support to students, faculty, patients, and doctors

PUBLICATIONS & RESEARCH

Ahmad Firas Ahmad Fer Rouse. "A Survey of Rogue Behavior in Autonomous AI." Submitted to ACM Computing Surveys (CSUR), 2025. [Under Review]

Ahmad Firas Azfar Ahmad Fer Rouse, Dr. Liang Cheng. "Interactive Graph-Based Reinforcement Learning for UAV Autonomy." Research Poster — USRCAP Undergraduate Research Program, University of Toledo, 2025.

TECHNICAL SKILLS

Languages & Dev: Python, SQL (PostgreSQL / PL-SQL), JavaScript, HTML/CSS, C++, Java, Embedded C, MATLAB, Bash/PowerShell

Enterprise & APIs: REST APIs, SOAP/Web Services, Power Automate, Azure AI Services, Microsoft Copilot Studio, Power Platform, Microsoft 365, SharePoint

Databases: PostgreSQL, MySQL, SQLite, MongoDB, AWS S3, pgAdmin 4 — RDBMS design, data governance, validation, disaster recovery

ML & AI: PyTorch, TensorFlow, scikit-learn, Reinforcement Learning (PPO, SAC, P3O), GNNs, Meta-Learning (MAML), NLP, Computer Vision, SHAP/LIME

MLOps & Data: Airflow, MLflow, Pandas, NumPy, Matplotlib, DOMO, Statistical Modeling, AWS (EC2, S3), XGBoost, Hyperopt

Dev Tools & SDLC: Git, Docker, CI/CD, Linux (Ubuntu CLI), LaTeX, CUDA, SSH, unit & system testing, technical documentation, code review, Agile

Robotics & Vision: ROS 2, OpenCV, YOLO11, NVIDIA Jetson Orin Nano, Sensor Fusion, RTOS, ZED Mini, OAK-D Pro

AWARDS & HONORS

- USRCAP Undergraduate Research Fellowship — \$3,000 grant, University of Toledo (2025)
- RocketHacks 2025 — Best Use of MongoDB Atlas
- AIChE Chem-E Car Competition — International Poster & Presentation, Most Innovative Car Design Worldwide
- AIChE Chem-E Car Competition — Regional Qualifier

PROFESSIONAL AFFILIATIONS

- IEEE — Institute of Electrical and Electronics Engineers (Member)
- AESS — Aerospace and Electronic Systems Society (Member)
- AIChE — American Institute of Chemical Engineers (Member)
- Pi Sigma Epsilon — Professional Sales & Marketing Organization (Member)